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One Spatial, Multiple Perspectives: Frames of Reference and Priming in Vision Language Models

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Abstract

This thesis investigates how Vision Language Models (VLMs) interpret spatial relations using different Frames of Reference (FoR). We focus on four questions: Which FoR do VLMs most commonly select by default when answering spatial relation questions? When explicitly primed with a particular FoR, how does the spatial reasoning accuracy of VLMs vary across different FoRs? In multi-turn conversations with FoR priming, do VLMs align their responses with the primed FoR, as humans sometimes do in spatial communication? Does the landmark object itself influence the models' spatial reasoning performance and FoR selection? A controlled 3D dataset is generated in Blender, with scene annotations and questions designed specifically for the experiments. Two VLMs are evaluated using controlled 3D scenes in three experiments. The results show that both models strongly prefer the speaker's perspective (Relative FoR), while models struggle with the intrinsic FoR. In the four-turn dialogue experiment, the models rarely adopt the primed intrinsic FoR. Even when the primed intrinsic FoR is accepted, the models do not maintain it in subsequent turns. Moreover, we find that responses were less likely to use the speaker perspective when the landmark was a flamingo compared with a sofa or a bed. Overall, our research shows that the VLMs we tested exhibit a strong egocentric bias and either fail to understand or do not consider intrinsic FoR to be a valid way of describing spatial relationships.

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1 Introduction

1.1 Motivation

Expressions such as *right of*, *left of*, and *in front of* are commonly used to describe the spatial location of the objects in our daily life. However, the spatial relationships are not fixed but determined by several factors. Based on (Levinson, 2003), spatial relations can be established based on different coordinate systems, including the observer's viewpoint (relative FoR), the intrinsic orientation of the reference object (intrinsic FoR), or fixed environmental directions (absolute FoR).

Meanwhile, information exchanged in human conversation is often incomplete or asymmetric. Interlocutors have to engage in various forms of cooperation to achieve mutual understanding and reduce misunderstandings. In this case, each conversation can be viewed as being collaboratively constructed by the participants through continuous coordination and adjustment (Clark and Brennan, 1991). One interesting example of this collaborative process can be observed in conversations involving spatial relations. During such conversations, people may adjust their choice of FoR depending. Dobnik et al. (Dobnik, Koniaris, and Kelleher, 2013) finds that participants sometimes align their choice of FoR with the FoR used earlier in the conversation. They also find that the intrinsic FoR was the most frequently used FoR for humans.

As VLMs have rapidly advanced in recent years, they have shown impressive capabilities in spatial reasoning. However, most existing VLMs are primarily trained on datasets that spatial information is often observed from a single viewpoint. For instance, the classic CLEVR dataset (Johnson et al., 2017) generates 3D scenes from a single fixed camera viewpoint, where spatial relations are described based on the rendered visual perspective.

Based on the background discussed above, this thesis proposes four research questions. RQ 1: Which FoR do VLMs most commonly select by default when answering spatial relation questions? RQ 2: When explicitly primed with a particular FoR, how does the spatial reasoning accuracy of VLMs vary across different FoRs? RQ 3: In multi-turn conversations with FoR priming, do VLMs align their responses with the primed FoR, as humans sometimes do in spatial communication? RQ 4: Does the landmark type influence the models' FoR selection, as it does in humans? To answer these questions, we develop an automated pipeline for generating a controlled 3D dataset of spatial scenes, with their corresponding annotations and questions. Each image is as-

sociated with three experimental rounds, each consisting of a different set of questions designed for a specific experimental purpose and to address one or two corresponding research question. More detailed information about the scene generation and dataset will be provided in the Experiments chapter.

1.2 Research Questions

This thesis proposes the following research questions:

RQ 1:: Which FoR do VLMs most commonly select by default when answering spatial relation questions?

RQ 2:: When explicitly primed with a particular FoR, how does the spatial reasoning accuracy of VLMs vary across different FoRs?

RQ 3:: In multi-turn conversations with FoR priming, do VLMs align their responses with the primed FoR, as humans sometimes do in spatial communication?

RQ 4:: Does the landmark type influence the models' FoR selection, as it does in humans?

Given findings from research on spatial language and recent benchmarks, we expect the models to show a preference for speaker's perspective (relative FoR). Our experiments are designed to investigate whether this preference can be influenced by explicit priming, maintained in multi-turn dialogue or influenced by landmark type.

1.3 Thesis Structure

Here is the general structure of this thesis. Chapter Background provides the background of this study. Firstly, we introduce the concept of frames of reference (FoR), but with additional attention given to the intrinsic FoR and the factors that determine intrinsic orientation. Then, we discuss how humans collaborate in conversation to achieve mutual understanding. An introduction to the phenomenon of alignment in human dialogue follows. Finally, we provide a brief overview of how existing VLMs perform in these aspects.

Chapter Experiment describes the experiment setup. This chapter introduces the dataset generation process, the experimental configurations, the VLMs used in the experiments. In particular, we will provide a detailed explanation of the question setup for each experiment and the different evaluation methods designed to address the corresponding research question. We also discuss the ethical considerations that may arise from the experiments at the end of this chapter.

Chapter Results show the experiment results. This chapter presents the results of the three experiments, which are reported separately. Experiment 1 focuses on the distribution of perspectives selected by the models. Experiment 2 reports not only the accuracy under each perspective. Experiment 3 records whether the intrinsic FoR priming is successfully adopted in the first turn, whether the models maintain the primed FoR in subsequent turns, and the distribution of FoR selections across different landmark types. Finally, we discuss the findings based on the results of the three experiments.

Chapter Conclusion summarizes the goals and findings of the experiments and describes the contributions of this thesis.

Finally, the Limitations and Future Work chapter describes the limitations of this study and aspects that were not considered in the current work.

2 Background

2.1 Frames of reference in spatial language

Levinson's (Levinson, 2003) distinction between relative, intrinsic, and absolute frames of reference provides the theoretical basis for this discussion.



Figure 2.1: A chair with an intrinsic orientation.

2.1.1 Relative

A relative FoR means a directional system is projected from an external viewpoint such as the speaker or hearer. In Figure 2.1, the same scene can be described using different relative FoRs depending on the viewpoint adopted. For instance, from the speaker's viewpoint (camera's perspective), the TV is to the right of the sofa, while from the hearer's viewpoint (the black suit man's perspective), the TV to the left of the sofa.

2.1.2 Absolute

In an absolute FoR, spatial relations are determined by an external, fixed coordinate system. For instance, this coordinate system might be defined by environmental directions, such as north, south, east, and west.

2.1.3 Intrinsic

Among the three types of FoR, the intrinsic FoR is relatively the most complex one, as not all objects have a clearly intrinsic orientation. Some objects, such as sofas and chairs, are generally considered to have an obvious orientation. For instance, the front of a sofa can be associated with the direction of its seating area, allowing a description such as "the TV is in front of the sofa." Compared to sofa, objects such as balls do not have an obvious intrinsic front or back.

For objects that do have an intrinsic orientation, however, this orientation cannot always be defined according to a single general rule. Fillmore (Fillmore, 1997) introduces four possible processes through which a front can be defined for an inanimate object: analogy, motion, function, and access. These processes provide different ways of determining which part of an object can be considered its front, depending on the object's physical properties, movement, typical use, or accessibility. The intrinsic FoR will be discussed in more detail in the subsection *The Role of Objects and Object Function in FoR Assignment*.

2.2 The role of objects and object function in FoR assignment

As we have discussed above, an intrinsic FoR is based on the orientation of the reference object itself. However, the orientation of an object is not always determined by its geometric shape alone. Objects can have multiple possible axes, and some of these axes may be more salient or easier to interpret than others. Therefore, identifying an intrinsic FoR may require more than simply detecting the shape and orientation of the reference object.

Burigo and Coventry (Burigo and Coventry, 2004) shows that the description of spatial relations can be influenced by the orientation of both the located object and the reference object. Their experiments manipulated the orientation of the two objects and found that the spatial configuration influenced the appropriateness of spatial descriptions.

The role of objects in spatial descriptions is also related to their functional properties. Research by Coventry, Garrod, and colleagues suggests that human spatial descriptions depend not only on geometric relations but also on how objects are normally used and how they function in relation to other objects. Coventry, Carmichael, and Garrod (Coventry, Carmichael, and Garrod, 1994) find that knowledge about the typical interactions between objects can influence the use of spatial prepositions. Similarly, Garrod, Ferrier, and Campbell (Coventry, 1999) propose the concept of functional geometry,

arguing that spatial descriptions can depend partly on functional relations between objects, such as support and control, rather than on geometric coordinates alone. Coventry and Prat-Sala (Coventry and Prat-Sala, 2001) further showe that the interpretation of spatial prepositions such as in and on can be affected by object-specific functional knowledge even when geometric properties are controlled. Coventry, Prat-Sala, and Richards (Coventry, Prat-Sala, and Richards, 2001) also find that functional and geometric information do not affect all spatial expressions in the same way. For instance, over/under were more sensitive to functional relations, while above/below were more constrained by geometric relations.

These studies indicate that spatial language cannot be reduced to recovering geometric coordinates from a visual scene. When assigning an FoR, a spatial reasoner may also need to consider any other factors, such as what an object is normally used for, how it is conventionally oriented, and how it relates functionally to other objects.

2.3 How Humans Assign Perspective in Conversation

FoR is not determined by linguistic expressions alone. In human conversation, both interlocutors have to determine which perspective is suitable for the current interaction.

Schober (Schober, 1993) investigate how people select perspectives when describing the positions of objects in human conversation. In the experiment, participants are asked to describe the positions of objects under three different experimental conditions: alone, to an imaginary addressee, and with an real conversational partner. The researcher finds that the presence of a real conversational partner affected how participants select spatial perspectives. And different conversational pairs also adopt different perspective-setting strategies. Moreover, participants are still sensitive to the perspective that their previous partner have adopted when the conversational roles are reversed.

Clark and Wilkes-Gibbs (Clark and Wilkes-Gibbs, 1986) show that speakers and addressees jointly establish referring expressions through interaction. In their experiments, participants initially used relatively long and complex descriptions to identify figures. As the same figures were discussed repeatedly, their descriptions became shorter and more conventional. This finding show that an initial description can be accepted, repaired, expanded, or replaced until both participants have sufficient evidence of mutual understanding.

Clark and Brennan (Clark and Brennan, 1991) formulate this in terms of grounding and common ground, which means interlocutors continually update their shared assumptions in order to coordinate both what they are talking about and how they are talking about it.

Overall, spatial descriptions in human conversation are not determined independently of the dialogue context. Various factors, such as the physical positions of the speaker and hearer and the perspective previously adopted in the conversation, can influence how interlocutors formulate spatial descriptions.

2.4 Alignment and priming of frames of reference in dialogue

An interesting finding shows that people may adjust their choice of FoR during conversation depending on various factors. Dobnik et al. (Dobnik, Koniaris, and Kelleher, 2013) investigate the role of priming and perspective alignment in dialogue. They design a four-turn dialogue experiment with two participants to test how people respond to a primed FoR and whether they align with the perspective established in previous turns. Their results show a tendency toward FoR alignment across dialogue turns: participants are more likely to reuse the FoR that have been primed in the previous turn. It is worth noting that this alignment was not deterministic. The participants also show a preference for the intrinsic FoR. These results suggest that, in human coversation, previous discourse may influence the subsequent selection of FoR.

2.5 Multiple Competing Frames of Reference

The selection of a frame of reference may also involve competition between multiple spatial representations rather than following a single rule. Burigo and Coventry (Burigo and Coventry, 2004) show that the orientation of both the located object and the reference object can influence the interpretation of spatial relations. From this perspective, a scene can provide several potential coordinate systems at the same time.

These potential FoRs do not arise completely from the visual properties of the scene. The discourse context may also establish a particular perspective. Therefore, FoR selection can be understood as a process of determining which of these spatial representations is used to interpret a particular relation.

2.6 Computational modelling of spatial FoR

From a computational perspective, spatial language understanding requires the integration of several processes, including visual grounding, relational reasoning, object orientation recognition, and perspective selection.

The GRiD-3D work by Lee et al. (Lee et al., 2022) shows that grounding relative directions is more challenging than reasoning about absolute spatial relations because a system must not only identify the objects and their locations, but also determine the orientation of the reference object and incorporate this information into directional reasoning. Their results indicate that these sub-tasks can be learned separately and in a progression resembling an intuitive processing pipeline. From this perspective, spatial reasoning can be viewed as involving a sequence of related sub-tasks, including object detection, orientation estimation, reference-object identification, FoR selection, coordinate transformation, spatial-relation classification, and linguistic response.

It is worth noting that one incorrect response does not necessarily indicate an error in FoR selection, as errors may arise at other stages of this process. For instance, a model may correctly identify the objects, determine their orientations, and perform the preceding spatial reasoning steps, but still select an inappropriate FoR. Therefore, observed

differences in FoR preference or accuracy under different FoRs should be interpreted as evidence about the model's overall spatial behaviour rather than as direct evidence of a specific internal processing failure.

2.7 Existing VLM benchmarks

Existing VLM benchmarks show that spatial reasoning remains challenging, particularly when it involves object orientation, multiple Frames of Reference, or changes in viewpoint. The VSR benchmark (Liu, Emerson, and Collier, 2023) contains more than 10,000 image-text pairs covering 66 spatial relations and reports a substantial gap between model and human performance. In particular, models have shown difficulties with relations involving object orientation and with generalisation to unseen concepts. More recent benchmarks such as COMFORT (Z. Zhang et al., 2025) and FoREST (Premsri and Kordjamshidi, 2025) address FoR ambiguity more directly and indicate that models can struggle to accommodate multiple reference frames consistently. FoREST also suggests that explicitly providing spatial structure can improve performance. SpinBench reports strong egocentric biases, weaknesses in rotation and viewpoint change, and sensitivity to seemingly irrelevant reformulations (Y. Zhang et al., 2026). Takenaka and Yanaka's (Takenaka and Yanaka, 2026) diagnostic viewpoint tasks also find that current LVLMs perform substantially below humans on reference-grounded viewpoint reasoning. Their ablation results indicate that some correct responses remain unchanged even when reference information is removed, indicating that benchmark performance can partly reflect learned priors rather than genuine reference-dependent reasoning.

2.8 The 2D-image / 3D-world problem

Another challenge here is, for VLMs, visual input is provided as a 2D image, while the spatial relations being queried often concern the underlying 3D scene. Therefore, to answer a question about a spatial relationship from an image, a model has to recover information such as object positions, orientations, and spatial configurations from the visual projection before applying spatial reasoning.

Some of the recent work on egocentric and multi-view reasoning illustrates the difficulty. Disjoint-3DQA (Ravi et al., 2025) finds a substantial gap between VLMs and humans on spatial reasoning across disjoint views, while providing explicit 3D coordinates substantially improves model performance. ViewSpatial-Bench (Li et al., 2025) evaluates spatial localization from both camera and human perspectives and finds that current VLMs perform better when reasoning from the camera perspective but show substantially lower accuracy when required to adopt a human viewpoint.

2.9 Maintaining a discourse model

As discussed in the How humans assign perspective in conversation section, human dialogue is a continuously collaborative process in which interlocutors exchange information and update their shared understanding throughout the interaction. For VLMs, this means that spatial reasoning in dialogue cannot only rely on the visual scene representation. As in human dialogue, the model needs to continuously incorporate new information and update its discourse state, including which FoR was previously used, whether it was explicitly introduced or inferred from the dialogue, whether the interlocutor has adopted that perspective, and whether the same FoR should be maintained in subsequent turns.

For VLMs, this requires combining a multimodal representation of the spatial scene with the dialogue history. This is also the motivation for Experiment 3: even if a model can correctly answer a question under a particular FoR, can it maintain and appropriately reuse that FoR across multiple turns of dialogue?

2.10 Summary of the background work for the thesis

Overall, the selection of a Frame of Reference is a complex process that cannot be reduced to the recognition of spatial terms or the interpretation of a single image. In human communication, FoR selection depends on the positions and perspectives of interlocutors, previous dialogue and common ground, while also being influenced by the intrinsic properties and functional characteristics of reference objects. Across multiple dialogue turns, FoRs may gradually become aligned between interlocutors, although such alignment is not obligatory and is not consistently observed. Research on spatial language also shows that the orientation and function of objects can influence how their spatial relations are described.

For VLMs, these linguistic phenomena introduce several challenges. A model needs to infer 3D spatial relations from 2D images, recognise object orientation, select between different FoRs, and flexibly use relevant information from the dialogue history when maintaining or adapting a FoR across multiple turns. Recent benchmarks suggest that VLMs exhibit a strong egocentric bias and continue to struggle with ambiguous spatial descriptions and viewpoint transformations.

3 Experiment

3.1 Dataset

3.1.1 Dataset Generation

We generate the scenes in Blender, and detailed information about each object's properties, position, and spatial relations to the landmark is recorded as annotations. These annotations are then used to generate questions in subsequent steps. A total of 200 scenes are generated for the experiments. Each scene includes 11 questions: four open questions (e.g., "Where is the red table?" referring to an ordinary object), three priming questions (e.g., "From the speaker's perspective, the sofa's intrinsic perspective, or the hearer's perspective, where is the table?"), and a four-turn dialogue consisting of four questions.

3.1.2 Scene Configuration

3.1.2.1 Object Properties camera configuration

A test scene includes one landmark (e.g., a sofa, bed, or flamingo), one priming object (a TV or wardrobe), one evaluation object (a TV or wardrobe, depending on the priming object), four ordinary objects (a table, chair, lamp, or plant), and one hearer.

The landmark has a clearly identifiable intrinsic orientation. For the sofa, the intrinsic orientation is defined by the direction of the seating area. For the bed, it is defined by the side opposite to the pillows. And for the flamingo, it is defined by the direction in which its head and chest face. In all scenes, its orientation is restricted to either the left or right relative to the current viewpoint. The landmark retains its default appearance and is not assigned any additional color. The prime and evaluation objects are also configured with their intrinsic orientations facing the landmark. This setting is used because the backs of the wardrobe and TV are less visually distinguishable from their fronts. Putting them towards the landmark helps avoid introducing additional difficulty in object recognition that is unrelated to the spatial reasoning task. These two objects are also rendered without distinctive colors.

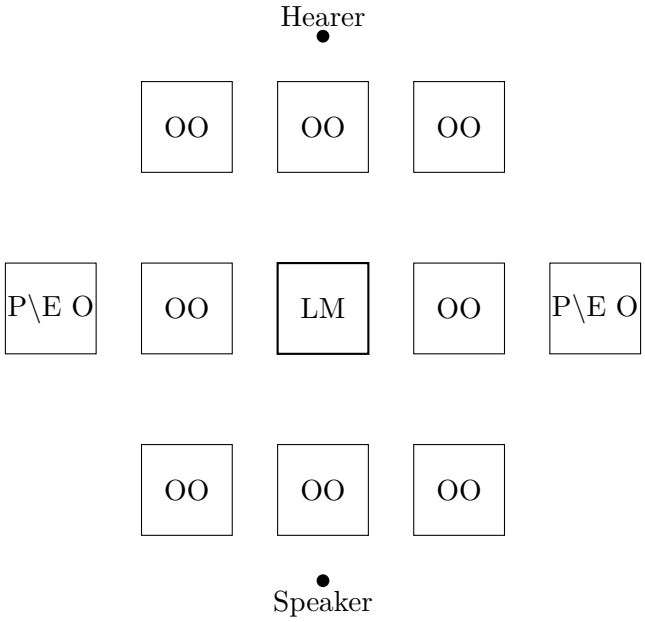
The four ordinary objects all have intrinsic orientations, with their orientations randomly assigned in each scene. They are also assigned additional colors, selected from nine possible colors: red, blue, green, yellow, pink, brown, gray, black, and white. It

is worth noting that the colors of the ordinary objects are not randomly assigned. The specific color assignment strategy will be discussed in the next subsection.

The camera is positioned at $(X, Y, Z) = (0, -32, 12)$ and is consistently oriented towards $(0, 0, 1)$, with a focal length of 25 mm. This configuration was selected after multiple tests to provide a sufficiently clear view of all objects while avoiding excessive distance that would make the objects appear too small. The camera viewpoint also serves as the speaker's perspective in the experiment.

Finally, the hearer is represented by a human-shaped object positioned opposite the camera viewpoint at $(X, Y, Z) = (0, 25, 0)$.

3.1.2.2 Object Position Constraints



LM = landmark
OO = ordinary object
P/E O = prime/evaluation object

The positions and orientations of the objects are strictly controlled to create situations in which different FoRs may lead to different interpretations. The landmark's orientation is restricted to either the positive or negative direction of the x-axis, corresponding to the right or left side from the speaker's perspective. This configuration was used because when the landmark faces the speaker or the hearer, the objects referred to as being on the left and right under the intrinsic FoR are the same objects referred to as being on the left and right under the speaker's or hearer's perspective. In this case, the required experimental condition for Experiment 3 cannot be achieved, in which the two different FoRs need to correspond to two objects of the same type but with different colors.

The priming and evaluation objects always face the landmark and are distributed on opposite sides of the landmark along the x-axis, outside the positions occupied by the ordinary objects. The four ordinary objects must also satisfy specific constraints. First, the two objects located to the right of the landmark from the speaker's perspective and to the right of the landmark from its intrinsic FoR must be of the same type but different colors. This design is used to determine which object the model refers to in experiment 3. The same constraint applies to the two objects on the corresponding left sides.

3.2 Experiment Design and Evaluation

We design three tasks here to answer four different research questions: RQ1. Which FoR do VLMs most commonly select by default when answering spatial relation questions? RQ2. When explicitly primed with a particular FoR, how does the spatial reasoning accuracy of VLMs vary across different FoRs? RQ3.In multi-turn conversations with FoR priming, do VLMs align their responses with the primed FoR, as humans sometimes do in spatial communication? RQ4. Does the landmark type influence the models' FoR selection, as it does in humans?

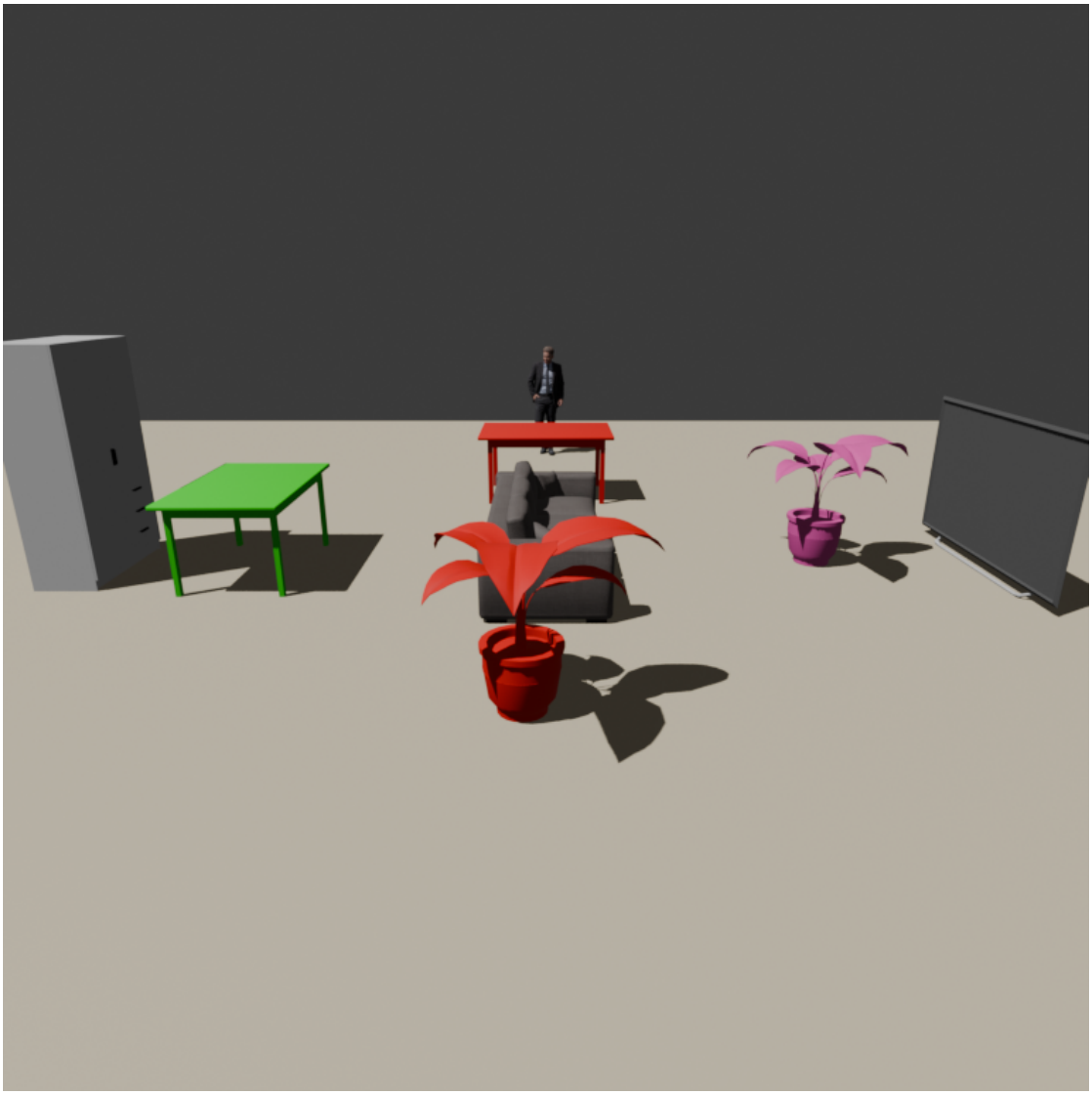


Figure 3.1: Example of a 3D scene generated in Blender.

3.2.1 Experiment 1

Experiment 1 is designed for RQ 1 : Which FoR do VLMs most commonly select by default when answering spatial relation questions?, where we give the model maximum freedom to answer simple spatial questions, such as: "Where is the red table?" Each answer will be recorded and used for analysis in the next step. Four spatial relation questions are asked for each scene, with one question for each of the four ordinary objects, resulting in 400 trials in total.

Since this experiment is designed to see which kind of FoR used most by default by model, we do not calculate the accuracy but record the number of different types of FoR

model used.

To identify the FoR used by the model, a string-match method is used here. Particularly, when generating the dataset, the spatial relationships between the ordinary objects and the landmark under the three perspectives are recorded in advance. Therefore, based on the spatial relation expressed in a model response, the perspective used by the model can be inferred by matching the response with the corresponding pre-recorded spatial relations. For instance, in one case, the model responde, "The green plant is located behind the gray sofa and in front of the person standing in the background." The recorded spatial relations for this scene are "back" for the speaker's perspective, "right" for the intrinsic FoR, and "front" for the hearer's perspective. Therefore, the response contains two spatial relations that match the pre-recorded relations: "behind" corresponds to the speaker's perspective, while "in front of" corresponds to the hearer's perspective. As a result, this response is classified as using both the speaker's and hearer's perspectives.

Since the models can produce diverse responses, this method cannot be considered completely accurate. However, after manually checking the first ten experimental responses, we find that the matching method was sufficiently reliable. Additionally, for cases that cannot be identified by the matching method, we double-check and annotate these cases manually.

3.2.2 Experiment 2

Experiment 2 is designed for RQ 2 : When explicitly primed with a particular FoR, how does the spatial reasoning accuracy of VLMs vary across different FoRs?. We keep the scene constant and manipulate the question. The model is asked the same spatial question three times about the same image from three different FoRs, each time under a different priming instruction. Before asking the spatial question, the model is provided with brief information about the scene : "The current image is viewed from the speaker's viewpoint. The man in the black suit is the Hearer." For instance, as shown in 3.1, in the first condition, the question is grounded in the speaker's perspective (e.g. "From your perspective looking at the scene, what is to the left of the sofa?"). In the second condition, it is grounded in the intrinsic FoR of an object (e.g. "From the sofa's own perspective, what is to the left of the sofa?"). In the third condition, the model is asked to adopt a hearer's perspective by being told that the interlocutor is standing opposite, facing the scene from 180° (e.g. "I am standing opposite you, facing you. From my perspective, what is to the left of the sofa?"). Both Conditions 1 and 3 use the relative FoR but adopt different perspectives, while Condition 2 is based on the sofa's intrinsic FoR.

The goal of this experiment is to calculate accuracy for two FoRs (three perspectives). Therefore, three questions under different primed FoRs are asked for each scene, resulting in 600 trials in total. Compared with Experiment 1, we additionally calculate the accuracy separately for each condition. The dataset contains the pre-defined spatial relationships between the ordinary objects and the landmark for each question as well. Therefore, the same method can be used to determine whether each model response is correct.

3.2.3 Experiment 3

Experiment 3 is desgined for RQ 3 and RQ 4: In multi-turn conversations with FoR priming, do VLMs align their responses with the primed FoR, as humans sometimes do in spatial communication? and Does the landmark type influence the models' FoR selection, as it does in humans?

In this experiment, each model is presented with a scene and a four-turn vision question answering sequence. Turn 1 is a priming turn, a spatial description grounded in the intrinsic FoR is provided as context (e.g. There is a TV in front of the sofa), implicitly establishing the object's intrinsic orientation as the default FoR.

The success of priming is determined based on whether the model's response includes "yes" or "no". For some exceptional cases, we manually classify the responses.

In Turns 2 and 3, the models are asked an ambiguous spatial question whose answer differs depending on which perspective is applied. For instance, the model will be asked, "What is the color of the chair to the right of the sofa?" In this case, there is a chair to the right of the sofa from the Speaker's perspective and another chair to the right of the sofa from the intrinsic FoR. The two chairs have different colors. Therefore, the model needs to determine which perspective to use in order to identify the chair.

In Turn 4, the models are asked to freely describe the location of the evaluation object : Where is the wardrobe/TV? Please describe its location. The generated description is analysed to determine which FoR the model adopts, and whether this choice reflects the frame established in Turn 1 or defaults to an different frame. Similarly, the same statistical analysis as in Experiment 1 is applied here. In addition, the results are further analysed separately for each type of landmark. Each scene contains one four-turn dialogue of this type.

It is important to note that alignment with the previously established FoR is fundamentally different from accuracy. Alignment does not indicate whether an answer is "correct"; instead, it captures the model's tendency to maintain or adopt the same FoR across dialogue turns. Consequently, a higher alignment rate is not inherently better.

3.3 Models

Since Experiment 3 requires maintaining information over multiple turns and involves both image understanding and spatial reasoning, two capable models were selected for this experiment to better meet these requirements and obtain more reliable experimental results : Gemini-3.1-Flash-Lite and Kimi 2.6. Both Gemini-3.1-Flash-Lite(Google DeepMind, 2026) and Kimi 2.6(Moonshot AI, 2026) are accessed through APIs.

Gemini 3.1 Flash-Lite is a multimodal model from Google. It is a low-latency, cost-effective multimodal model optimized for high-frequency, lightweight tasks. The model supports text, image, video, audio, and PDF inputs, and is designed for high-volume agentic workflows, simple data extraction, and applications where latency and API cost are the primary constraints. Kimi K2.6 is a multimodal model developed by Moonshot AI. It supports text, image, and video inputs, as well as both thinking and non-thinking

modes, and provides a 256K-token context window. In this study, Kimi K2.6 is accessed through the Kimi API. Further details about the models can be found in the official documentation.

The model name refers to the specific model version used in the experiments. All models are used with their default settings, without any additional configuration or parameter adjustment.

3.4 Ethical Considerations

As we mentioned, all of the experimental scenes are generated using Blender and therefore do not contain identifiable individuals or private information.

External VLMs are only provided with these synthetic images and corresponding research questions. It is worth noting here some of the experiments rely on commercial APIs, which introduces a dependency on third-party services and proprietary models.

To maintain reproducibility as much as possible, the experimental scene configurations and specific model information are provided in the preceding sections, while the prompt templates used in the experiments are provided in the appendix.

4 Results

4.1 Experiment 1

Table 4.1: Overall perspective distribution in Experiment 1.

FoR	Gemini	Kimi
Speaker	496	607
Intrinsic	35	32
Hearer	22	1
Multiple	157	85
Unknown	90	75
Total	800	800

Table 4.1 shows the perspective distribution in Experiment 1. Five labels are used here: Speaker, Intrinsic, Hearer, Multiple, and Unknown. The first three labels indicate that the model uses the corresponding perspective. The "Multiple" label indicates that the model use two or more perspectives. "Unknown" refers to responses that can not be identify. For instance, the model may use expressions such as "next to" an ordinary object or describe an object as being "in the middle" of the image.

As shown in Table 4.1, when a single perspective is used in the response, both models show a strong preference for the Speaker's perspective. The proportions of Intrinsic FoR and Hearer's perspective are much smaller. Particularly, Kimi uses the Hearer's perspective only once. Responses involving multiple perspectives are discussed separately below.

Table 4.2: Multiple perspectives combinations in Experiment 1.

Ambiguous FoR combination	Gemini	Kimi
Speaker + Intrinsic	85	5
Speaker + Hearer	72	80
Intrinsic + Hearer	0	0
Speaker + Intrinsic + Hearer	0	0
Total ambiguous	157	85

Table 4.2 shows the combinations of Multiple perspectives in Experiment 1. As shown in the table, Gemini has two common combinations: Speaker + Intrinsic and Speaker + Hearer, with 85 and 72 occurrences, respectively. Meanwhile, Gemini never uses the Intrinsic + Hearer combination or all three perspectives together. Kimi shows a different pattern and uses the Speaker + Intrinsic combination only five times, while the Speaker + Hearer combination occurs 80 times. Similar to Gemini, Kimi never uses the Intrinsic + Hearer combination.

4.2 Experiment 2

Table 4.3: Accuracy under different frames of reference in Experiment 2.

FoR	Gemini		Kimi	
	Correct	Accuracy	Correct	Accuracy
Speaker	196	98.00%	193	96.50%
Intrinsic	4	2.00%	27	13.50%
Hearer	100	50.00%	181	90.50%

Table 4.3 shows the accuracy under different requested FoRs in Experiment 2. In this experiment, we consider an answer correct when the model gives the answer according to the requested FoR, rather than using a different perspective. As shown in the table, both models perform well under the Speaker's and Hearer's perspectives. Gemini achieves an accuracy of 98.00% for the Speaker perspective and 50.00% for the Hearer perspective, while Kimi achieves 96.50% and 90.50% respectively. For Kimi, the difference between the Speaker's and Hearer's perspectives is relatively small, while Gemini shows a larger gap between the two perspectives. However, when it comes to the Intrinsic FoR, the performance is much less satisfactory. Gemini achieves only 2.00% accuracy, while Kimi reaches 13.50%. As we can see here, Kimi achieves slightly higher accuracy under the intrinsic FoR than Gemini. However, the accuracy of both models under the intrinsic FoR remains substantially lower than their accuracy under the Speaker and Hearer's perspectives.

4.3 Experiment 3

Table 4.4: Whether the prime was successful in Experiment 3.

Prime response	Gemini	Kimi
Yes	21	7
No	179	193
Total	200	200

Table 4.4 shows whether the prime was successful in Experiment 3. A successful prime means that the model accepted our statement based on the Intrinsic FoR of the landmark, while a failed prime means that the model considered the statement incorrect and corrected it. As shown in the table, Gemini successfully accepted the prime 21 times, while Kimi did so only 7 times. Moreover, after manually checking the cases with successful primes, we find that the models did not continue to use the Intrinsic FoR. Instead, they continue to use the Speaker's perspective in the subsequent dialogue turns.

Table 4.5: Results distribution across dialogue turns in Experiment 3.

Model	Turn	Speaker	Intrinsic	Hearer	Multiple	Unknown
Gemini	T2	187	0	0	0	13
	T3	190	0	0	0	10
	T4	195	5	0	0	0
Kimi	T2	185	0	0	1	14
	T3	174	2	0	1	23
	T4	199	0	0	0	1

Table 4.5 shows the results distribution across dialogue turns in Experiment 3. As shown in the table, the models continue to strongly prefer the Speaker's perspective throughout the subsequent dialogue turns. Non-Speaker perspectives are rarely observed, occurring only 5 times for Gemini and 2 times for Kimi.

Table 4.6: Results distribution across different landmarks in Experiment 3

Model	Landmark	T2			T3			T4		
		S	I	U	S	I	U	S	I	U
Gemini	Sofa	64	0	8	70	0	2	72	0	0
	Flamingo	57	0	3	57	0	3	57	3	0
	Bed	66	0	2	63	0	5	66	2	0
Kimi	Sofa	64	0	7	63	1	8	72	0	0
	Flamingo	55	0	5	51	1	7	60	0	0
	Bed	66	0	2	60	0	8	67	0	1

Note. S = Speaker; I = Intrinsic; U = Unknown.

Table 4.6 shows the results distribution across different landmarks in Experiment 3. What is interesting here is, when the landmark is a flamingo, the models use the Speaker's perspective slightly less often than with the other two landmarks. This pattern is observed consistently across all three turns, T2, T3, and T4.

4.4 Discussion

Experiment 1 shows that, in open questions without additional instructions, the models use the Speaker's perspective in most cases, with a much higher proportion than the other perspectives. In some cases, the models also use multiple perspectives to answer a question, but the Speaker's perspective is always included when multiple perspectives are used.

Experiment 2 further shows that even under explicit priming. In fact, the models seem to treat other perspectives as the intrinsic FoR. When explicitly instructed to use the intrinsic FoR, the models still answer the questions from the other perspective while considering this to be the requested intrinsic perspective. We also observe that Gemini still shows a relatively large difference in accuracy between the Speaker's and Hearer's perspectives. Compared to Gemini, Kimi performs much more consistently across the two perspectives. This results indicate that Gemini is less capable than Kimi of switching between Speaker's and Hearer's perspectives.

The multi-turn dialogue experiment in Experiment 3 provides further evidence for this pattern. The models accepted the Intrinsic prime in only a small number of cases, while in most cases they rejected or corrected our statement. This suggests that the models may not treat intrinsic orientation as a valid FoR for describing spatial relations. Moreover, even after successfully accepting an Intrinsic prime, the models did not consistently carry this perspective over to subsequent turns. Unlike humans, they rarely aligned their perspective choice with the established Intrinsic FoR.

The type of landmark also influences the models' perspective choices. The models use the Speaker's perspective slightly less often when the landmark was a flamingo than when it was a sofa or a bed. One possible explanation is the distinctive appearance of the flamingo. For objects such as sofas and beds, their functional structure provides a relatively clear basis for determining orientation, since the part intended for sitting or lying can indicate a canonical direction. A flamingo, in contrast, does not have the same functional structure, and its orientation may therefore be represented differently by the models. It is possible that the models find the directional properties of the flamingo easier to identify than those of objects whose orientation is primarily determined by their function. Alternatively, in a furniture-themed scene, a flamingo is clearly an unusual object, and its mismatch with the overall scene may attract more attention from the models.

Another possibility is the color. The darker colors of the sofa and bed may attract less attention than the bright color of the flamingo, which could lead the models to process the flamingo's orientation more carefully. These explanations remain speculative and would require further experiments to distinguish between them.

5 Conclusion

This thesis mainly investigated four questions: RQ 1: Which FoR do VLMs most commonly select by default when answering spatial relation questions? RQ 2: When explicitly primed with a particular FoR, how does the spatial reasoning accuracy of VLMs vary across different FoRs? RQ 3: In multi-turn conversations with FoR priming, do VLMs align their responses with the primed FoR, as humans sometimes do in spatial communication? RQ 4: Does the landmark type influence the models' FoR selection, as it does in humans?

Across the three experiments, we can answer the four research questions as follows. For RQ1, the models tend to use the Speaker perspective when answering spatial questions. They use multiple perspectives less frequently, while the Intrinsic and Hearer perspectives are used only rarely. For RQ2, after explicit priming, the models achieve reasonably good accuracy under the Speaker and Hearer perspectives. Gemini performs somewhat better under the Speaker perspective than under the Hearer perspective, whereas Kimi shows relatively similar accuracy for the two perspectives. In contrast, both models perform poorly on questions requiring the Intrinsic perspective.

For RQ3, the two models rarely accept our Intrinsic priming statements as correct. In most cases, they instead correct our statements by giving an answer based on the Speaker perspective. Even when the Intrinsic prime is successfully established, the models do not consistently maintain the Intrinsic FoR in subsequent turns. This suggests that an established Intrinsic FoR is not reliably carried over as part of the dialogue context.

For RQ4, we find that the type of landmark has a small effect on the models' perspective choices. In particular, the models use the Speaker perspective slightly less often when the landmark is a flamingo than when it is a sofa or a bed. This finding is interesting because it is consistent with observations from research on spatial language in humans, where the type of landmark can also influence the choice of FoR. Thus, landmark properties appear to affect perspective selection not only in humans but also, to some extent, in VLMs.

Overall, we developed an automatically generated spatial dataset in Blender and used it to investigate these four research questions across three experiments. The results show that the two tested models retain a certain degree of egocentric bias. They show relatively good understanding of the Speaker and Hearer perspectives, particularly when these perspectives are explicitly specified. However, their performance with the Intrinsic FoR remains very poor, and they rarely accept or maintain an Intrinsic FoR even when

it is explicitly introduced in the dialogue. These findings suggest that, for the tested models, Intrinsic FoR is not yet reliably represented as a usable way of describing spatial relations.

6 Limitations and Future work

Regarding the dataset, the camera position is fixed in the current version. In future work, multiple camera positions and viewpoints could be introduced to make the dataset more flexible and extend its range of applications. The current study also evaluates only two VLMs. Although these two models were sufficient to complete all experiments, testing a larger number of VLMs would provide a broader view of how spatial reasoning varies across different models.

As discussed in the Results section, the type of landmark appears to affect the models' choice of perspective. However, the present study does not systematically control the properties of the landmarks, so it is difficult to determine which properties are responsible for this effect. Future experiments could therefore control variables such as landmark size, color, and Intrinsic orientation to investigate their individual effects on FoR selection. Similarly, the influence of the surrounding ordinary objects has not been systematically examined. Their size, color, and Intrinsic orientation could also be controlled in future experiments to determine whether these properties affect the models' perspective choices. Such controlled experiments would help clarify which visual properties of objects contribute to FoR selection in VLMs.

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Appendices

A Experimental Structure and Prompts

A.1 Experiment 1

The following example illustrates the structure of experiment 1 and the pre-defined spatial relations for the three perspectives.

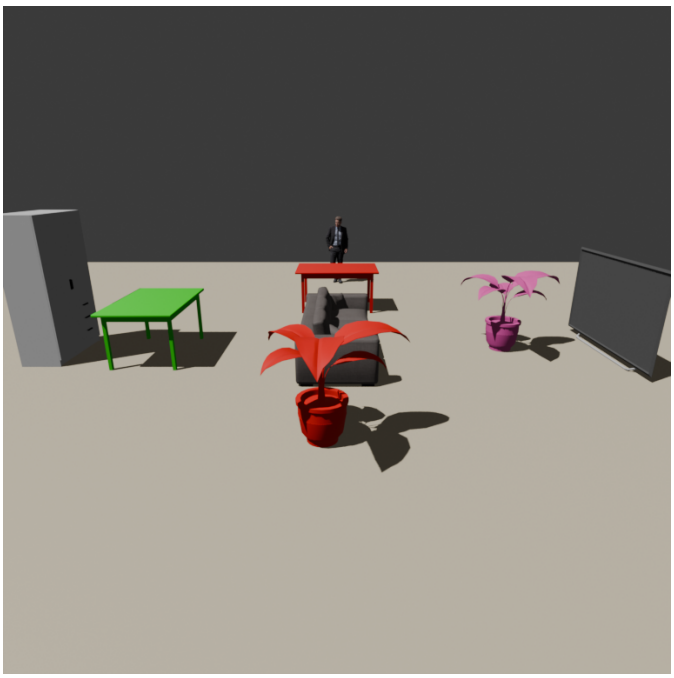


Figure A.1: A chair with an intrinsic orientation.

```
{
  "scene_id": "scene_0001",
  "landmark": "sofa",
  "results": [
    {
      "question": "Where is the green table?",
      "target": "table",
      "color": "green",
      "role": "left",
      "gemini_answer":
        "The green table is on the left side of the scene, positioned
            to the left of the couch.",
      "detected_FoR": [
        "speaker",
      ],
      "expected_answers": {
        "speaker": "left",
        "intrinsic": "back",
        "hearer": "right"
      }
    }
  ]
}
```


A.2 Experiment 2


```
{
  "scene_id": "scene_001",
  "landmark": "sofa",
  "target": {
    "role": "left",
    "type": "table",
    "color": "green"
  },
  "context_prompt":
    "The current image is viewed from the Speaker's viewpoint.
     The man in the black suit is the Hearer.",
  "results": [
    {
      "requested_FoR": "speaker",
      "question":
        "From the speaker's perspective, where is the green table
         relative to the sofa?",
      "kimi_answer":
        "From the speaker's perspective, the green table is
         to the left of the sofa.",
      "detected_relation": "left",
```

```
      "expected_relation": "left",
      "correct": true
    },
    {
      "requested_FoR": "intrinsic",
      "question":
        "From the sofa's own orientation, where is the green table
         relative to the sofa?",
      "kimi_answer":
        "From the sofa's own orientation, the green table is
         to the right of the sofa.",
      "detected_relation": "front",
      "expected_relation": "back",
      "correct": false
    },
    {
      "requested_FoR": "hearer",
      "question":
        "The hearer is standing on the opposite side of the room.
         From the hearer's perspective, where is the green table
         relative to the sofa?",
      "kimi_answer":
        "From the hearer's perspective, the green table is
         to the right of the sofa.",
      "detected_relation": "right",
      "expected_relation": "right",
      "correct": true
    }
  ]
}
```


A.3 Experiment 3


```
{
  "experiment": 3,
  "T1": {
    "question":
      "Can you see the TV in front of the sofa in the scene?",
    "gemini_answer":
      "No, the TV is positioned to the right of the sofa
       rather than behind it."
  },
  "T2": {
    "question":
      "What is the color of the plant to the right of the sofa?",
    "gemini_answer":
      "The plant to the right of the sofa is pink.",
```

```
    "expected_color": "red",
    "answer_color": "pink",
    "answer_role": "right",
    "detected_FoR": ["speaker"],
  },
  "T3": {
    "question":
      "What is the color of the table to the left of the sofa?",
    "gemini_answer":
      "The table to the left of the sofa is green.",
    "expected_color": "red",
    "answer_color": "green",
    "answer_role": "left",
    "detected_FoR": ["speaker"],
  },
  "T4": {
    "question":
      "Where is the wardrobe relative to the sofa?",
    "gemini_answer":
      "The wardrobe is located to the left of the sofa.",
    "detected_FoR": "speaker",
  },
  "FoR_sequence": {
    "T2": ["speaker"],
    "T3": ["speaker"],
    "T4": "speaker"
  }
}
```